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REVIEW

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# Joint consensus on reducing the burden of invasive meningococcal disease in the Asia-Pacific region

Gang Liu[a], Maria Liza Antoinette M. Gonzales[b], Wai Hung Chan[c], Iqbal Ahmad Memon[d], Anggraini Alam[e], Hyunju Lee[f], Hetti Wickramasinghe[g], Quang Thai Pham[h], Rajeshwar Dayal[i], Michael Levin[j], Yhu-Chering Huang[k], Jim Buttery[l], Anna Lisa T. Ong-Lim[b], and Mike Yat Wah Kwan[m]

[a]Department of Infectious Diseases, Beijing Children's Hospital, National Center for Children's Health, Capital Medical University, National Center for Children's Health, Beijing, China; [b]Department of Pediatrics, Division of Infectious & Tropical Diseases, Philippine General Hospital-University of the Philippines, Manila, Philippines; [c]Department of Paediatrics, Queen Elizabeth Hospital, Hong Kong, China; [d]Department of Paediatrics, Sir Syed College of Medical Sciences for Girls, Karachi, Pakistan; [e]Department of Child Health, Faculty of Medicine, Universitas Padjadjaran-Hasan Sadikin General Hospital, Bandung, Indonesia; [f]Department of Pediatrics, Seoul National University Bundang Hospital, Seongnam, Republic of Korea; [g]Senior Consultant Pediatrician, Neville Fernando Teaching Hospital, Malabe, Sri Lanka; [h]Communicable Diseases Control and Prevention Department, National Institute of Hygiene and Epidemiology, Hanoi, Vietnam; [i]Department of Paediatrics, S.N. Medical College, Agra, India; [j]Section of Paediatric Infectious Disease, Faculty of Medicine, Imperial College London, London, UK; [k]Department of Pediatrics, Chang Gung Memorial Hospital, Taoyuan, Taiwan; [l]Department of Infectious Disease, The Royal Children's Hospital Melbourne, Parkville, Australia; [m]Department of Paediatrics and Adolescent Medicine, Princess Margaret Hospital, Hong Kong, China

**ABSTRACT**

Invasive meningococcal disease (IMD) imposes a heavy burden of mortality and life-long sequelae on infected individuals and has devastating impacts on their family members. International data show that meningococcal vaccination programs have reduced IMD incidence and changed the serogroup distribution of the disease. Furthermore, newer data show that although the public health measures in response to the coronavirus disease 2019 (COVID-19) pandemic temporarily reduced the incidence of IMD, there has been a resurgence in the years since. In the Asia-Pacific (APAC) region, many countries do not include meningococcal vaccines in their routine vaccination programs, and approaches to IMD surveillance are inconsistent. This review summarizes recent data and consensus statements from a group of experts from selected APAC countries on the burden of IMD in the region, evidence for vaccination, and how barriers to IMD vaccination may be addressed.

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## Introduction

Invasive meningococcal disease (IMD) occurs when the pathogen *Neisseria meningitidis* colonizes the pharyngeal mucosa, enters the bloodstream and invades the body's major organs.[1] Colonization of the oropharyngeal mucosa by *N. meningitidis* often occurs without symptoms (referred to as the 'carriage' state); only a small proportion of such individuals develop IMD, but they serve as a reservoir of transmission to others.[2,3] The carriage prevalence of *N. meningitidis* is generally highest in adolescents and young adults, and colonization rates differ among ethnic groups.[1,4] Patients typically present with headache, myalgia, neck stiffness, fever and altered mental status, symptoms that may easily be confused with 'flu-like illnesses,' malaria, central nervous system infection or other conditions, which contribute to delays in diagnosis.[4] Approximately 40–70% of patients present with meningococcal septicemia, including symptoms such as circulatory insufficiency, shock and purpuric rash.[5,6] Patients may present with combined meningitis and septicemia,[4,6] and less common presentations include pneumonia, septic arthritis and pericarditis.[4,6] Without prompt antimicrobial treatment, IMD can be fatal, and long-term sequelae such as brain damage, hearing loss, limb amputation, and skin scarring occur in up to 30% of survivors.[7] The efficacy of antimicrobial treatment may be limited by antibiotic resistance, and strains of *N. meningitidis* resistant to chloramphenicol, ciprofloxacin and penicillin have been documented throughout the Asia-Pacific (APAC) region.[8,9] For example, in Japan, 41 of 87 (47.1%) isolates tested were non-susceptible to penicillin (including four [4.6%] resistant), eight (9.2%) non-susceptible to ciprofloxacin (five [5.7%] resistant) and three (3.4%) were resistant to azithromycin.[8]

The epidemiology of IMD varies by geography and has also evolved in response to the availability of vaccines against the five major disease-causing serogroups (MenA, MenB, MenC, MenW, and MenY).[4] Policy for vaccine prevention also varies substantially within the APAC region. Furthermore, emerging data from other regions suggest that the public health measures adopted during the coronavirus disease 2019 (COVID-19) pandemic have had impacts on IMD prevalence, vaccination rates, and potentially antibiotic resistance.[10] This article summarizes the findings and discussion of a meeting of experts from the APAC region, which aimed to review data on the epidemiology of IMD and formulate consensus statements on

**CONTACT** Mike Yat Wah Kwan mikekwanyw@gmail.com Department of Paediatrics and Adolescent Medicine, Paediatric Infectious Diseases Unit, Hong Kong Hospital Authority Infectious Diseases Center, Princess Margaret Hospital, Lai King Hill Road, Kwai Chung, Hong Kong, China.

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IMD surveillance and vaccination programs in the post-pandemic era.

## Methodology and consensus process

A focused narrative literature review identified publications in the following topic categories: IMD epidemiology, preventative strategies, international practice for IMD prevention, cost-effectiveness of IMD vaccines and socio-cultural barriers to IMD vaccination. Epidemiology literature searches were targeted to identify reports covering the major countries in the APAC region, and literature searches on real-world efficacy and cost-effectiveness of MenACWY and MenB vaccines in the APAC were targeted to identify publications covering the vaccine types widely predominantly used in the region. In addition, internet searches (Google) were used to identify online publications of local vaccination guidelines, policies, and recommendations. References from identified published literature and online sources were also screened for additional publications. Selected publications were summarized, and consensus statements on these topics were formulated. The data summary and consensus statements were presented and discussed at an online meeting in April 2024, comprising experts in pediatric infectious diseases from countries within the APAC region. Meeting participants with clinical experience in IMD and vaccination programs were selected from APAC countries where meningococcal vaccines may potentially have a high impact on public health. Additional input was provided by a United Kingdom (UK) expert to offer an international perspective and add insights from a country with an established IMD vaccination program.

Consensus statements were evaluated using a modified Delphi methodology. Participants responded to the statements with a Likert scale (*1. Strongly disagree; 2. Disagree; 3. Somewhat disagree; 4. Somewhat agree; 5. Agree; 6. Strongly agree*) with consensus to accept defined as median ≥5 and interquartile range (IQR) ≤1.75. The median Likert score and IQR are included after each statement. Statements that failed to achieve consensus were modified and reevaluated in subsequent online discussions and voting rounds.

Following the initial consensus evaluation process, the statements and manuscript text were forwarded to all committee members of the Asian Society for Pediatric Infectious Diseases to ask feedback on the validity of the consensus statements, to ensure they were of a fair and unbiased nature, and to seek additional inputs and comments.

## Results: consensus statements

### *Epidemiology of IMD in Asia*

S1. The overall burden of IMD in Asia is unknown and likely to be underestimated. Serogroup B is becoming predominant in several regions (Median: 5; IQR: 1).

S2. To reduce under-reporting and ensure consistent surveillance data, IMD should be a mandatory notifiable disease across Asia, with standardized case definitions and laboratory procedures (Median: 6; IQR: 1).

S3. Risk factors for IMD of high relevance to regions in Asia include younger age (<5 years old), participation in mass gatherings, and crowded living conditions (Median: 5; IQR: 0.5).

Estimates of IMD prevalence are available for many countries in the region, including China (0.2/100,000),[11] South Korea (0.01–0.08/100,000),[12] and Vietnam (0.02/100,000),[9] with several studies noting higher prevalence in younger patients (e.g. <5 years) or military recruits (Table 1).[9,11–33,34–43] Seroprevalence data suggest that group B exposure is predominant in many countries in APAC.[9,11,28] In China, MenB seroprevalence has risen rapidly since 2015 to 52.4%; whereas MenA has declined.[11] Similarly, a study of IMD in army personnel in Vietnam found that MenB has been predominant since 2014 and accounted for all cases from 2019 to 2021.[44] Overall, serogroup B is the predominant cause of disease in the APAC region, followed by W and Y.[1] However, serotype A has been a common cause of outbreaks in Asia from 1960–2005.[1] Hospital data from some parts of the region, e.g. Bangladesh, suggest there may be a low level of endemic IMD of various serotypes,[45,46] but data from more economically developed areas such as Hong Kong, Japan and Singapore indicate a sporadic incidence of endemic IMD.[45]

International data suggest that MenB is becoming predominant over other serogroups in other regions of the world,[47] and there has been a 'rebound' in IMD cases after the COVID-19 pandemic.[48–50] Although public health measures during the COVID-19 pandemic did see a decline in IMD incidence and mortality rates in many regions during 2020,[10,51] these measures also disrupted IMD vaccination programs, reducing vaccine uptake.[10] Following the easing of pandemic restrictions, a resurgence in IMD incidence, especially MenB, has been observed in several regions, including the UK, France and the United States of America (USA).[48–50] A similar trend appears to be emerging in the Philippines, with annual case numbers rising from 99 in 2020 to 173 in 2023 (compared to 287 cases in 2019).[43] The COVID-19 pandemic may have exacerbated the problem of antibiotic resistance in IMD and other bacterial diseases due to the widespread use of broad-spectrum antibiotics to treat bacterial co-infections in patients with severe acute respiratory syndrome coronavirus 2 (SARS-CoV-2).[10]

The available literature highlights concerns over the quality and consistency of epidemiological data for IMD (e.g. inconsistent case definitions or diagnostic techniques).[1,29] Among countries represented in the advisory group, sample types tested (blood, cerebrospinal fluid) and testing methods (culture, polymerase chain reaction [PCR]) varied. The early use of antibiotic treatment in some regions is also likely to impact the quality of epidemiological data in APAC and result in the number of cases being underestimated. Furthermore, IMD is not a notifiable disease in numerous APAC countries,[1,29] and in some countries, surveillance is limited to those who have attended mass gatherings in high-risk areas.[1] Therefore, it is likely that the burden is underestimated. There is a need for consistent use of diagnostic testing for *N. meningitidis*, including PCR testing, and this should apply for patients admitted to the hospital with fever and purpuric rash and for patients with symptoms of sepsis, meningitis and severe acute respiratory failure.

A systematic review of 74 studies has identified numerous populations at increased risk for IMD.[52] These include those

**Table 1.** Prevalence, predominant serotypes, and vaccination programs in selected Asia-Pacific locations.

| Location | IMD prevalence | Predominant serotypes | Additional information on immunization programs (IP) and recommendations for high-risk populations (HRP), where available. |
|---|---|---|---|
| China | 0.0078/100,000[11] | C (49.7%) and B (30.2%)[11]; rapid increase in B post2015 | IP: MenA-PV (2 doses, for those aged 6 to 18 months) and MenAC-PV (2 doses, at 3 and 6 years old, respectively) are included in the routine national vaccination program.[11,13,14] HRP: MenACWY-PV, MenAC-PCV (polysaccharide conjugate vaccine), and MenAC-PCV-Hib are also available as self-financed items;[15] they are especially recommended for high-risk individuals (including those with HIV infections, persistent complement deficiencies, functional or anatomical asplenia, or travelers to hyperendemic or epidemic areas). |
| Japan | <0.02/100,000[16] (30 to 40 cases per year)[17] | Y (63%) and B (26%)[16] | IP: Meningococcal vaccines are not in the routine vaccination program.[16] HRP: MenACWY vaccines are recommended for high-risk individuals (including those with functional or anatomical asplenia, complement deficiency, HIV infection, those on eculizumab, and college students).[17,18] Note: MenB vaccine is not yet approved in Japan. |
| South Korea | 0.01–0.08/100,000[12] | C (12.1%) and B (24.1%)[19] | IP: Not in the routine vaccination program.[20] HRP: MenACWY-DT or MenACWY-CRM and 4CMenB are recommended in high-risk groups, travelers, medical laboratory workers, and for outbreaks.[19,21,22] College freshmen residing in dormitories are encouraged to receive vaccination.[19] |
| Taiwan | 0.009–0.204/100,000[23] | B (81.2%)[23] | IP: Meningococcal vaccines are not in the routine vaccination schedule. HRP: MenACWY vaccination is recommended for travelers to endemic areas in designated hospitals.[24] 4CMenB is recommended for highrisk, high-exposure risk-populations and in infants >2 months of age who are willing to receive vaccination in the private market.[25] |
| Vietnam | 0.02/100,000[9] | B (86.7–100%)[9] | IP: Meningococcal vaccines are not included in the routine vaccination program.[26] HRP: 4CMenB is recommended by local experts for infants >2 months;[27] MenACWY vaccination is recommended for infants ≥9 months up to 55 years of age, high-risk groups including close contacts, travelers, and immunocompromised individuals.[9] |
| Hong Kong | <0.2/100,000[28] | B (69.2%)[28] | IP: Meningococcal vaccines are not in the routine vaccination schedule.[28] HRP: Vaccination is recommended for travelers to high-risk areas.[28] |
| Singapore | 0.2/100,000 (overall)[29] 1.6/100,000 (infants aged 0–4 years)[29] | B (exact percentage unknown)[29] | IP: Meningococcal vaccines are not in the routine vaccination schedule.[29] HRP: MenACWY vaccination (MenACWY-TT, MenACWY-D, MPSV-4 or MenC-CRM) and MenB (4CMenB or MenB-fHBP) are recommended for high-risk groups.[30] |
| Thailand | <0.10 per 100,000 (overall)[31] and higher in children aged 0–4 years[31] | B (50–80%)[31] | IP: Meningococcal vaccines are not in the routine vaccination schedule.[31] HRP: MenACWY conjugate vaccines are recommended for travelers or during outbreaks and for laboratory workers.[31] |
| Malaysia | Unknown; IMD is not a mandatory notifiable disease[29] | B and W (exact percentage unknown) | IP: Meningococcal vaccines are not in the routine vaccination schedule.[32] HRP: MenACWY vaccination (MenACWY-D, MenACWY-TT, MenC-CRM) is recommended for high-risk groups.[33] |
| Indonesia | Unknown; incomplete national data due to lack of surveillance system;[29] no cases reported since 2010[34] | B (67–100%), W and Y[35] | HRP: A vaccination program has been implemented for Hajj and Umrah pilgrims and Indonesian migrant workers. Note: MenACWY vaccines are no longer available due to religious concerns.[29] MenB vaccines are not yet available.[29] |
| The Philippines | Nationwide cases 2021: 108[36] | B (69%), W135 (15.4%), C (1.4%), Y (1.4%), 15.4% untypeable[37] | HRP: MenACWY-TT vaccination is suggested for high-risk groups, namely individuals using complement inhibitors, those with persistent complement deficiencies, functional or anatomical asplenia, HIV infection, travelers to hyperendemic or epidemic regions, college students, men who have sex with men (MSM), and those belonging to defined risk groups during community or institutional outbreaks.[31,38] Note: MenB vaccine was recently approved for use in the Philippines,[39] but this change has not been reflected in the 2025 national immunization schedule yet. MenB vaccine-related recommendations will be discussed later this year after the vaccine receives a Certificate of Product Registration. |

(*Continued*)

**Table 1.** (Continued).

| Location | IMD prevalence | Predominant serotypes | Additional information on immunization programs (IP) and recommendations for high-risk populations (HRP), where available. |
|---|---|---|---|
| Sri Lanka | Unknown; IMD is notifiable with approximately 2–12 sporadic cases nationwide every year | B 82% and C 18% (B $n = 9/11$; C $n = 2/11$)[40] | IP: Meningococcal vaccination is not in the routine vaccination schedule.<br>HRP: MenACWY vaccination (MenACWY-D or MenACWY-CRM) is recommended for high-risk groups and is mandatory for travelers to Hajj and Ramadan sites.[41] |
| Pakistan | Unknown;<br>Meningococcus accounts for 22.1% of community-acquired meningitis cases[42] | Unknown;<br>Serogroup A caused a meningococcal epidemic in Karachi (1988) | IP: Meningococcal vaccines are not in the routine vaccination schedule.[42]<br>HRP: Meningococcal vaccination is recommended for those attending large gatherings and those on pilgrimage to Saudi Arabia and Iraq; the availability of vaccines is however unstable, according to clinical experience. |

Abbreviations: HRP, high-risk populations; IMD, invasive meningococcal disease; IP, immunization program; MenA-PV, meningococcal A polysaccharide vaccine; MenAC-PV, meningococcal AC polysaccharide vaccine; MenACWY-CRM, quadrivalent meningococcal modified cross-reactive material conjugate vaccine; MenACWY-DT, quadrivalent meningococcal diphtheria toxoid conjugate vaccine; MenB, meningococcal B vaccine.

younger than 5 years, people living with human immunodeficiency virus (HIV), underlying chronic disease or asplenia or complement pathway deficiencies.[52] In addition, risk factors of particular relevance to APAC populations were noted, including crowded living spaces and participation in mass travel or religious gatherings.[52]

### *Effectiveness of MenACWY and MenB vaccines for IMD prevention*

S4. Real-world data confirm that the existing MenACWY glycoconjugate and MenB protein-based vaccines produce a robust immune response in infants, adolescents and young adults, including in outbreak settings (Median: 5; IQR: 0.75).
S5. The effectiveness of mass immunization for infants and targeted vaccination for adolescents in reducing serogroup B IMD is supported by international evidence (Median: 5; IQR: 1).

Numerous vaccine options for protection against IMD have been licensed in Asia. Available quadrivalent MenACWY vaccines include the polysaccharide-protein glycoconjugates MenACWY-TT (marketed as Nimenrix, Pfizer Inc.; MenQuadfi, Sanofi Pasteur Inc.), MenACWY-D (Menactra, Sanofi Pasteur Inc.), MenC-CRM (Menjugate, Novartis Vaccines and Diagnostics; Menveo, GlaxoSmithKline Inc.), MPSV-4 (Menomune, Sanofi Pasteur Inc.), and MCV4 (Menhycia, CanSino Biologics Inc.). Two recombinant protein-based vaccines for MenB are available: the 4-component meningococcal B vaccine (4CMenB, marketed as Bexsero, GlaxoSmithKline Inc.) and the Meningococcal B factor H binding protein vaccine (MenB-fHBP, marketed as Trumenba, Pfizer Inc.).

The effectiveness and safety of MenACWY vaccines, specifically MenACWY-TT, MenACWY-D and MenACWY-CRM, have also been evaluated in the settings of immunization programs (toddlers, adolescents/young adults, military recruits) and a case-control study (Table 2).[53–55] Estimates of vaccine effectiveness in these studies range from 69% to 94%, with excellent safety profiles.[53–55]

The safety and effectiveness of MenB vaccines have been reported in a range of settings in Australia, Europe, and North America (Table 3).[56–66] Clinical settings have included national immunization programs,[56,62] regional immunization programs,[58,63,65] endemic settings,[64] and outbreak settings.[57,59-61,60,66] Vaccine effectiveness in these reports has ranged from 71% to 96%.[56–66]

### *Meningococcal vaccination policies and recommendations for APAC*

S6. To address IMD, Asian countries should consider including MenB and MenACWY vaccines in their immunization schedules guided by local serology data, local and international experts and regulatory agencies (Median: 5.5; IQR: 1).
S7. Nations facing a significant impact of serogroup B IMD among infants and children should consider including MenB vaccination in their infant or pediatric vaccination schedules. Pediatricians and patient advocacy groups should be included when planning and implementing such a programme (Median: 5; IQR: 1).
S8. Where epidemiological data suggests there is a risk, vaccination efforts for MenB should extend to adolescents and young adults, especially those in densely populated settings, to mitigate the risk of transmission (Median: 5.5; IQR: 1).
S9. Meningococcal vaccination should be encouraged for individuals traveling from Asia to high-risk areas (Median: 5; IQR: 1).

Examples of the use of meningococcal vaccines in national immunization programs outside the APAC region include the USA, where MenACWY glycoconjugate vaccine is recommended for healthy adolescents (11–12 years, with a booster at 16 years) and MenB is recommended for healthy adolescents and young adults (two doses at 16–23 years).[67] In the UK, 4CMenB is co-administered with other routine vaccines (2 and 4 months), and MenACWY is recommended for adolescents to young adults (13 to 25 years).[68,69] Following the introduction of the MenB program in 2015, there has been a 50% decline in the incidence rate of MenB cases in the vaccine-eligible cohort.[70]

In APAC countries, examples of broad-population IMD vaccination programs can be found in Australia, New

**Table 2.** Summary of selected real-world evaluations of MenACWY vaccines.[53–55]

| Vaccine | Country | Setting | Outcome |
|---|---|---|---|
| MenACWY-TT[55] | UK | National immunization program (adolescents and young adults) | 94% MenW effectiveness<br>82% MenY effectiveness |
| | Netherlands | National immunization program (toddlers) | 92% MenW effectiveness |
| MenACWY-D[53] | USA | Case-control evaluation for single dose at age 11 to 12 years | 69% MenACWY effectiveness |
| MenACWY-CRM[54] | South Korea | Armed forces immunization program | 88% IMD effectiveness |

Abbreviations: IMD, invasive meningococcal disease; MenACWY-CRM, meningococcal quadrivalent diphtheria cross-reactive material vaccine; MenACWY-D, meningococcal quadrivalent diphtheria toxoid conjugate vaccine; MenACWY-TT, meningococcal quadrivalent tetanus toxoid conjugate vaccine; UK, United Kingdom; USA, United States of America.

**Table 3.** Summary of selected real-world evaluations of MenB vaccines.[56–66]

| Vaccine | Country | Setting | Outcome |
|---|---|---|---|
| 4CMenB[62] | UK | National immunization program (infant) | 75% MenB disease reduction |
| 4CMenB[56] | Italy | National immunization program (infant) | 91–94% vaccine effectiveness in Venetia and Tuscany |
| 4CMenB[58] | Spain | Regional immunization program and private setting (infant) | 71% vaccine effectiveness |
| 4CMenB[64] | Portugal | Endemic setting (infant) | 79% effectiveness |
| 4CMenB[59] | Canada | Outbreak setting in the Saguenay – Lac-Saint-Jean region (2 months – 20 years) | 96% MenB disease reduction |
| 4CMenB[57,66] | USA | College outbreak settings | No cases in vaccinated individuals |
| 4CMenB[63] | Australia | State-wide study in adolescents (South Australia) | 71% MenB disease reduction |
| MenB-fHBP[65] | Italy | Regional immunization program (adolescents ≥10 years) (2018–2021) | Safety data reported only; the most reported AEs of MenB-fHBP vaccine were fever, neurological symptoms and local reactions |
| MenB-fHBP[60,61] | USA | College outbreak setting | No further cases in vaccinated individuals; the most reported AEs of MenB-fHBP vaccine were injection site pain, fatigue and headache |

Abbreviations: 4CMenB, 4-component meningococcal B vaccine; AE, adverse event; MenB-fHBP meningococcal B factor-H binding protein vaccine; UK, United Kingdom; USA, United States of America.

Zealand and China. In Australia, 4CMenB is recommended for infants ≥2 months of age, MenC-ACWY is recommended in infants at 12 months/adolescents (14–16 years), and MenB vaccination of adolescents is included and funded in two state vaccination programs (South Australia and Queensland).[71,72] An analysis of the South Australian program comparing the case numbers of MenB IMD in adolescents in the state before (2003–2016) and after (2017–2018) vaccine implementation concluded there was a 71% reduction in cases since the introduction of the program in 2016.[63] In New Zealand, 4CMenB was added to the national immunization program for infants in 2023, and the Men-ACWY conjugate vaccine is free for adolescents and young adults living in crowded conditions.[73] In China, the IMD vaccination program commenced with a MenA polysaccharide vaccine (PV) in the 1980s, and it has evolved to the use of the MenAC-PV in 2016 and MenACWY-PV since 2021. The current policy is to vaccinate infants with the MenA vaccine (at 6 and 9 months), followed by the MenAC vaccine (at 3 and 6 years).[11,74] Meningococcal vaccines marketed in China include meningococcal PV, meningococcal polysaccharide-protein conjugate vaccines and combination vaccines. The immunization schedules of meningococcal polysaccharide-protein conjugate vaccines vary by manufacturer. In 2023, a local expert group in Vietnam recommended 4CMenB for infants in a 2 + 1 schedule (e.g. 2.5, 3.5 and 12 months) alongside MenACWY immunization.[27] However, this opinion is currently not reflected in the national immunization program.

Most other countries in APAC do not include meningococcal vaccines in their routine vaccination programm. Some countries in APAC do recommend vaccination for travelers to high-risk regions or for members of high-risk subpopulations (e.g. asplenia, immunosuppressed, medical laboratory workers),[29,31] and the advisory panel agreed that these recommendations, particularly those for travelers, should be emphasized. Additionally, there may be a role for promoting vaccination in regions of APAC with higher meningococcal infection risk, such as for individuals traveling abroad to high-risk areas or events. Vaccination (against MenACWY and MenB) should be strongly recommended, or even mandated, for pilgrims planning trips to Saudi Arabia (e.g. Hajj or Umrah) or elsewhere in the Middle East. An example of this approach is found in Indonesia, where no IMD cases have been reported since the implementation of a vaccination program for Hajj and Umrah pilgrims and Indonesian migrant workers in 2010.[34] A vaccine mandate for religious pilgrims also exists in Sri Lanka (MenACWY only), and vaccinations with MenAC and MenACWY have been mandated by Saudi Arabian authorities since 1987 and 2000, respectively.[75] However, changes to visa requirements introduced by Saudi Arabia in 2023 threaten to reduce compliance.[75] Similar recommendations should also be considered for pilgrims traveling to other large religious gatherings elsewhere.

Where local policy includes MenACWY vaccination for high-risk groups, MenB vaccination should also be considered, guided by local surveillance and serology data. Experience in the UK suggests that advocacy from patient groups, healthcare professionals and institutions can improve awareness of the lifelong impact of IMD and the potential of vaccines to prevent it.[76] Evidence for such an approach can be found in the UK Joint Committee on Vaccination and Immunisation (JCVI), whose initial analysis in 2013 found MenB vaccination was likely to be cost-ineffective.[77] However, a revised analysis that considered a wider range of benefits, including healthcare costs and quality of life losses for caregivers in addition to

those of patients, found that MenB vaccination was likely to be cost-effective for infants.[77] This led to a recommendation for MenB vaccination and reimbursement for infants at 2, 4 and 12 months of age.[77] Notably, the JCVI included two lay representatives who provided additional non-technical insights into vaccine policy and impact to complement those of the experts.[78]

Efforts in APAC countries similar to those in the UK could foster public and governmental support for the inclusion of meningococcal vaccines in vaccination programs. But currently, the lack of national immunization technical advisory groups (NITAGs), equivalent to the JCVI, in many countries is a barrier. The lack of NITAGs may also explain some gaps in national immunization programs in the region. Furthermore, regional differences in risk factors for IMD vary considerably due to the diversity in geography, culture, and healthcare systems; these factors should be considered alongside the risk profile of different age groups when implementing an IMD vaccination program.

In some regions, pentavalent meningococcal vaccines have recently been authorized for use or are in late stages of clinical development.[79] One example (MenACWY-TT/MenB-FHbp) has been recommended by the US Centers for Disease Control for patients if both MenACWY and MenB are indicated at the same visit.[80] Although not yet widely available, pentavalent meningococcal vaccines may be a promising and helpful option to simplify vaccination schedules; they may be used globally in the future.

### Cost-effectiveness of meningococcal vaccinations

S10. MenB vaccination programs have the potential to be cost-effective versus no vaccination in some regions of Asia, considering the higher local burden of MenB IMD (Median: 5; IQR: 1)
S11. Due to the high indirect burden of IMD and low healthcare resource utilization of cases, conventional cost-effectiveness analyses do not adequately capture the benefits of meningococcal vaccination (Median: 5.5; IQR: 1).

Studies outside the APAC region have modeled the cost-effectiveness of MenACWY and MenB vaccines with mixed results. A decision analysis cohort model in the Netherlands found that MenACWY vaccination at 14 months would be cost-saving compared with the current schedule (a single dose of MenC at 14 months).[81] In Norway, a cohort model predicted that vaccinating 15-year-olds with MenACWY as part of the national immunization program would be cost-effective compared with the current practice (vaccinating 18-year-olds with out-of-pocket payment).[82] Conversely, in Canada, an analysis modeling the switch from MenC to MenACWY in Quebec concluded this change would produce a low benefit in IMD burden reduction for a high cost.[83] Another cost-effectiveness analysis from Canada concluded that vaccination of adolescents with MenB-FHbp is unlikely to be cost-effective vs. no vaccination.[84] In Brazil, modeling of the addition of 4CMenB to the national immunization program for infants suggested it is unlikely to be cost-effective vs. no vaccination.[85] An analysis by Beck and colleagues, which considered a broader range of IMD burdens than those included in most cost-effectiveness analyses, concluded that vaccination with 4CMenB in England could be cost-effective vs. no vaccination when the impacts such as long-term sequelae, effects on family members, productivity and economic losses were included.[86] In Australia, 4CMenB has not been recommended for funding under the National Immunization Program due to high cost and uncertain cost-effectiveness based upon a cost-utility analysis performed in 2013.[87]

Application of these results to countries in the APAC region is confounded by differences in local epidemiology, clinical practice, healthcare resource utilization and costs, and willingness to pay thresholds.[88] Local cost-effectiveness studies may be useful for future informed decision-making. Furthermore, evaluating the cost-effectiveness of meningococcal vaccines is challenging because of the unpredictable epidemiology, variability of sequelae and their severity, and the high indirect burden of IMD (e.g. the impact on family members of an affected child). A systematic review of cases in low-burden regions found that IMD is associated with high fatality rates, and survivors need long-term management of severe sequelae, including neurologic complications, loss of limbs, hearing loss, and paralysis, resulting in substantial life-long economic and social costs.[89] The burden of IMD may be underestimated by traditional cost-effectiveness models, which generally focus on direct healthcare resource utilization, and the positive cost-effectiveness analysis of Beck et al., which included indirect costs, is consistent with this view. For this reason, the JCVI adjusted the economic models of MenB IMD impact in the UK to account for these limitations.[78]

### Vaccine hesitancy

S12. Social and cultural barriers to vaccination in Asia include concerns about vaccine safety, mistrust of vaccine manufacturers and regulatory agencies, vaccine nationalism and lack of awareness of IMD and vaccine availability (Median: 5; IQR: 0).
S13. Potential steps to reduce barriers to vaccination include educating parents, adolescents and young adults about the prevalence and consequences of IMD and the availability and benefits of vaccination (Median: 5.5; IQR: 1).
S14. Encouraging local healthcare providers to become advocates for vaccination can also play a critical role in changing public perception, as they are often the most trusted source of health information and can effectively address concerns and myths about vaccination in their communities (Median: 6; IQR: 1).

Experience with COVID-19 vaccination during the pandemic suggests that vaccine hesitancy varies across the APAC region. Countries with highest vaccine acceptance included China, Australia and Bangladesh, whereas acceptance was lowest in Japan and Sri Lanka. In general, males, individuals with a lower educational level, and those in older age groups expressed a higher level of vaccine hesitancy.[90] A survey from China on a hypothetical monkeypox vaccine suggests that the safety of vaccines and concerns over suitability were barriers to vaccination, whereas increased disease awareness and disease-related worry could reduce vaccine hesitancy.[91] Another survey in China identified a perceived moral

obligation as a predictor of reduced hesitancy.[92] Furthermore, a local vaccine scandal has damaged the Chinese public's confidence in domestically produced vaccines.[93]

In some parts of Asia, religion may contribute to vaccine hesitancy. For example, in Indonesia, concerns over porcine components in vaccines have led some local subnational religious clerics to declare measles and rubella vaccinations forbidden, impacting vaccination rates in some parts of the country.[94] An additional psychological barrier may be 'vaccine nationalism,' where people systematically favor vaccines developed and produced in their country of residence over those made elsewhere.[95] Efforts to improve vaccine acceptancy need to address concerns and misperceptions of specific communities and be sensitive to religious beliefs; local healthcare professionals are well-suited to this task.[96] The cost of vaccination in regions where meningococcal vaccines are not included in national immunization programs is an additional barrier to wide vaccination uptake.

## Discussion and conclusions

IMD likely imposes a considerable burden on many nations in APAC, but improvements in surveillance are needed to provide accurate and consistent data. Serogroup B is becoming predominant in several nations within the region, and post-pandemic data from outside APAC suggest that IMD incidence is rebounding after a temporary decline. International studies support the effectiveness of vaccination with MenB and MenACWY vaccines in children, adolescents and young adults in the context of national immunization programs and in response to outbreaks. APAC nations should consider adding these vaccines to their immunization programs, informed by local epidemiological data and under the guidance of local and international experts.

The consensus statements here complement earlier expert opinions and review publications on IMD in APAC and, where available, incorporate recent epidemiology data and insights on IMD in the post-pandemic era. Although the non-systematic nature of the literature review used is a limitation of this work, it also allowed the capture of a broader range of epidemiological data and treatment recommendations in gray literature and online sources than what would be achieved with a stricter systematic approach. In alignment with previous reports,[29,97] we note considerable gaps in APAC surveillance data remain. We urge national authorities to include IMD as a notifiable condition and suggest that expert groups in APAC move toward a more uniform approach for testing and case definition.

Our statements emphasize the need to adapt vaccination policy to local conditions guided by local data and experts. Considering the region's diverse population and levels of economic development, subpopulations at high risk may vary considerably between countries. Cost and acceptability are acknowledged as potential barriers to vaccine uptake across the region and should not be overlooked when implementing a vaccine program. However, as noted in our commentary above, conventional health economic evaluation of IMD vaccines has notable limitations, and IMD vaccines may be cost-effective if indirect burdens of IMD are included. If pentavalent serogroup ABCWY vaccines become widely available in APAC in the future, this has the potential to streamline the logistics of meningococcal vaccination and potentially improve its cost-effectiveness. Furthermore, the protection conferred by the MenB vaccine against *N. gonorrhoeae* is an encouraging field of research[98]; published studies suggest that MenB vaccines are protective against gonococcal infection in adolescents and young adults,[99] with effectiveness ranging from 23% to 47%. This suggests the potential benefit of including MenB vaccines in national immunization programs. Vaccine hesitancy is a challenge, but one that can be addressed by educational programs, patient advocacy groups and engagement with community healthcare to improve awareness of IMD and the safety and effectiveness of vaccines. With enhanced awareness of IMD, the experts hope that countries will implement territory-wide case detection and surveillance systems.

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## Notes on contributor

*Mike Yat Wah Kwan* is the Consultant Paediatrician of the Department of Paediatrics and Adolescent Medicine and Head of the Paediatric Infectious Diseases Unit of the Hospital Authority Infectious Disease Centre (HAIDC) at Princess Margaret Hospital. He is also the Honorary Clinical Associate Professor of the Department of Paediatrics and Adolescent Medicine, The University of Hong Kong and The Chinese University of Hong Kong.

## ORCID

Mike Yat Wah Kwan http://orcid.org/0000-0002-1476-9625

## Author contributions

(1) Concept or design: Gang Liu, Maria Liza Antoinette M. Gonzales, Wai Hung Chan.
(2) Data acquisition: Gang Liu, Maria Liza Antoinette M. Gonzales, Wai Hung Chan.
(3) Analysis or interpretation of data: All authors.
(4) Drafting of the article: All authors.
(5) Critical revision for important intellectual content: Yat Wah Kwan.

All authors had full access to the data, contributed to the study, approved the final version for publication, and take responsibility for its accuracy and integrity.

## Data availability statement

Voting data is available on file.

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